The Alien Bootstrap Hypothesis:
A Conditional Strategic Model of Transport-Limited Colonization

Gabriel Kent*

June 15, 2026

Abstract

The Alien Bootstrap Hypothesis (ABH) is a conditional strategic model of extraterrestrial colonization under transport constraints. It supposes that non-human intelligence can reach Earth but cannot move biological populations, ecosystems, or industrial infrastructure at colonization scale, and asks under what conditions waiting for the host civilization to develop local biological-instantiation capability dominates immediate transport. We argue that the waiting advantage is not a tunable parameter but a consequence of *knowledge locality*: the developmental viability of a novel Earth-adapted organism accrues only by running Earth development, so no degree of the visitors' advancement pre-solves it, and the visitors face the same developmental gate the host faces. ABH's distinctive contribution is that it *dates* the wait. Coupling the strategic premise to the Kent Equation's developmental clock $K_{\text{tech}}(t)$ converts an indefinite "they are watching" into a time-indexed, monitorable forecast whose single discriminating signature is a correlation between visitor behavior and $K_{\text{tech}}(t)$. We are explicit that the model is speculative and conditional: current official evidence does not establish extraterrestrial visitation, and ABH inherits that premise rather than supplying it. The value of the exercise is the disciplined conditional reasoning, not the truth of its assumptions. This is a companion to the Kent Equation paper [1], which supplies the mechanism ABH treats as a clock.

1 Introduction

Speculative models of extraterrestrial behavior routinely make an inference that does not follow: that a civilization able to cross interstellar distance must therefore be superior across the entire technology tree, and that arrival implies the capacity to do as it wishes on arrival. ABH rejects that inference. *Arrival capability and colonization capability are different capabilities.* A civilization might hold a narrow, perhaps physics-deep transport breakthrough while facing hard limits on mass movement, ecological transfer, industrial relocation, and biological viability in a foreign environment. Under such limits the rational path is not to transport bodies across interstellar distances. It is to wait until the host planet has independently built the substrate that allows bodies to be instantiated locally — converting colonization from a transport problem into a local fabrication problem.

Two analogies fix the intuition and recur below. The first is a *bell*: a craft whose propulsion depends on a field, resonance, or metric configuration may function only within a constrained physical envelope, so that adding mass, altering internal geometry, or carrying diverse biological

*Rhomega Corp., gabriel@rhomegacorp.com

and material loads changes the system enough to make it unreliable — much as modifying a bell changes its tone. Capability in such a craft would be real but *narrow.* The second is a *forest*: one can build a house alone in the wilderness, but life there is costly because everything must be done from scratch; it is cheaper to wait until a town and its infrastructure arise nearby and to build against that. The bell motivates the transport limit; the forest motivates the waiting advantage. Neither is offered as evidence.

We are explicit at the outset about evidentiary posture, because the subject invites overclaiming. The NASA Unidentified Anomalous Phenomena Independent Study Team found no conclusive evidence in the peer-reviewed literature that reported UAP are extraterrestrial in origin, and treated extraterrestrial explanation as the hypothesis of last resort [2]. The U.S. All-domain Anomaly Resolution Office's historical review reported no evidence that any government investigation or review panel has confirmed a UAP sighting as extraterrestrial, and no empirical evidence for claims of reverse-engineered extraterrestrial technology [3]. ABH does not contest this record. It does not claim its premises are true. It claims only that *if* they hold, a specific and disciplined consequence follows, and that the existing technical apparatus already names the instrument that times the consequence.

That instrument is the Kent Equation [1]. Its contribution to ABH, and the contribution that distinguishes ABH from every nearby framework, is not the waiting — waiting without contact is the zoo hypothesis [4] — but the *dating* of the waiting. Bolted onto the Kent clock, an indefinite "they are observing us" becomes a time-indexed forecast with a sharp and falsifiable-in-principle tell: track developmental biology, not genome synthesis. The paper is admittedly more philosophical than rigorous, because the subject is irreducibly speculative; but speculation is not an excuse to abandon valid reasoning, and the aim throughout is to keep the inference clean even where the premises cannot be secured.

2 Relation to the Kent Equation

The Kent Equation is *agent-neutral* and *premise-free.* It computes $K_{\text{tech}}(t) = F_{\text{dev}}(t \mid M_{[0,t]})$, the probability that a host civilization has crossed the developmental-viability threshold required to fabricate a viable biological body to specification on its own surface by time t, driven by a substrate-maturity index $M(t)$ built from energy, general-purpose computation, and biological experimental throughput [1]. It contains no claim about who might wish to occupy the bodies. The question *when can this world grow bodies?* is logically prior to, and separable from, the question *who wants them?*

ABH fills the premise slot the mechanism deliberately leaves empty. In one line:

> The Kent Equation supplies the clock; the Alien Bootstrap Hypothesis supplies the reason a transport-limited visitor would care about the clock.

This division is not cosmetic. It is what allows ABH to carry all of the project's speculative weight without contaminating the mechanism. Because τ_{dev} — the binding developmental gate — depends only on the host's substrate $M(t)$ and carries no agent-specific content, conditioning the mechanism on any application premise leaves its reading unchanged:

$$P\big(\tau_{\text{dev}} \le t \mid M_{[0,t]},\ \text{premise}\big) = P\big(\tau_{\text{dev}} \le t \mid M_{[0,t]}\big) = K_{\text{tech}}(t). \tag{1}$$

The premise selects *whether the clock is worth watching*; it does not alter the clock's reading. A reader who rejects everything in this paper loses the application but keeps the mechanism intact. That is by design.

3 The premise and its epistemic status

ABH rests on a conjunction of assumptions. We state them plainly and then, more usefully, sort them by what kind of claim each is, because a flat list of six co-equal suppositions is both less honest and easier to attack than an explicit account of which assumptions are load-bearing and which merely amplify.

The assumptions as stated.

A1. Extraterrestrial non-human intelligence exists.

A2. It has reached Earth recurrently for at least the modern era (roughly eight decades) while remaining concealed.

A3. The associated sightings indicate craft of minimal scale and minimal carrying capacity.

A4. Advanced transport does not imply advancement across the rest of the technology tree; the visitors appear narrow rather than uniformly superior.

A5. Bootstrap colonization is the visitors' primary mode of expansion — a biological analogue of terraforming.

A6. Premature or unambiguous self-revelation would tend to hinder the host's development of the very technologies the visitors require.

Sorting by epistemic status. These do not have equal standing, and saying so is the honest move.

Visitation (V, from A1–A2) is assumed, against the current official record. The eight-decade concealment is part of the premise — a supposition we adopt, not an observation we explain.

Limitation (L, from A3–A4) is also assumed, and is the more exposed of the two. There is no evidence about the payload of alien craft, because there is no confirmed alien craft. What the reports could supply, conditional on being genuine craft at all, is *consistency*: small craft with few occupants and a light footprint are *consistent with* transport limits. That is consistency, not evidence, and it is weak even as consistency, since the same reports fit "misidentification," "classified human technology," and "probes limited by design rather than by constraint" equally well. The bell (§1) is a conjecture about *why* transport might be narrow; it is not an observation that it is. We therefore place L on the same tier as V: assumed. Assumption A4 specifically — the narrowness of the visitors' wider technology — is treated below not as a load-bearer but as an *amplifier* (§4.3); the central result does not require it.

The waiting advantage (W) is the one term that is earned rather than assumed. It is derived in §4 from the locality of developmental knowledge, and that derivation, not any premise, is the analytical core of the paper.

The terraforming framing (A5) is a strong but useful framing assumption, discussed in §5. *The concealment premise (A6) is an assumption that pays its way* by converting an awkward fact into a prediction, discussed in §6.

The honest compression is therefore: **two assumed premises (V, L), one earned result (W), and three further suppositions (A4–A6) that shape or strengthen the picture without bearing its logical weight.** This is a smaller and more defensible structure than "six things we hope are all true," and we will be careful to flag, at each step, which assumption a given conclusion actually consumes.

4 The waiting advantage is grounded in knowledge locality

The hypothesis invites an immediate objection. If the visitors can cross interstellar distance, why would they wait on the maturation of a less advanced host's biotechnology? Stated baldly, W looks like special pleading: the visitors are conveniently advanced where they need to be (transport) and conveniently backward where the story needs them to be (biology).

The objection is dissolved not by adjusting parameters but by a structural fact about the kind of knowledge at issue. The argument runs in five steps, and consumes only V and L — not A4.

1. **The visitors need an Earth-adapted hybrid.** By hypothesis they are not adapted to Earth's biochemistry, atmosphere, microbial environment, or gravity. The probability that their unmodified genome yields an Earth-viable organism is vanishingly small. They require not merely a body but a designed hybrid: a genome authored to be viable in Earth conditions.

2. **Designing the hybrid requires knowing which specification is Earth-*viable*.** This is not the *synthesizability* of a sequence — computing a writable string that realizes an intended design, a computational question the visitors may well have solved. It is its *viability*: whether a given design will actually develop into a functioning organism in Earth's environment.

3. **Earth-viability is Earth-specific knowledge.** It depends on Earth biochemistry, developmental signaling, immunological context, and gestational environment. It is not derivable from physics and not transferable from the visitors' own biology. It is a fact about Earth developmental processes.

4. **Earth-viability knowledge accrues only by running Earth development.** This is the developmental bottleneck of the Kent paper restated for a novel organism: inference can design, but it cannot observe a developmental process that has not been run. For a never-before-existing hybrid, viability is confirmed only empirically.

5. **Therefore no degree of computational or transport advancement pre-solves the problem.** The visitors face the same τ_{dev} the host faces, because τ_{dev} is a property of running Earth development, not of who runs it.

This is the content of W, and it is not special pleading, because the visitors' advancement is simply in the wrong currency. Their compute may design synthesizable sequences and their technology may write them — plausibly long before arrival — but the two operations that demand Earth-developmental capability, viability-design and growing, both route through τ_{dev}. The visitors then have exactly two paths:

- independently rebuild Earth-developmental biology at scale, covertly and without the host's infrastructure — which is just to *become* a covert instantiation-capable presence, the same mechanism performed by a different agent; or
- wait for the host to mature τ_{dev} and make use of the result.

That the second is cheaper than the first is the whole of W. **The waiting advantage is not assumed; it follows from the locality of developmental knowledge.** We are careful about scope: this strengthens the hypothesis's internal coherence, not its truth. The premise V remains assumed.

4.1 The economic layer: why waiting dominates among the feasible strategies

Knowledge locality establishes that the visitors *cannot* pre-solve the developmental gate. It does not by itself rank the strategies that remain feasible once that is granted: a covert build-from-scratch presence, a limited local base plus patience, managed integration near the threshold, or pure waiting. Ranking these is an economic question, and here the forest analogy is exact. Waiting *externalizes the capital cost of the colonization scaffold* — energy, logistics, computation, automation, medicine, developmental biology, governance — onto the host, which builds it for its own reasons. The comparison can be written as an optimal-stopping or real-options condition: waiting until a later interval dominates immediate action when

$$\mathrm{EU}(\text{wait until } t+\Delta) \;>\; \mathrm{EU}(\text{colonize now}), \tag{2}$$

an inequality that becomes more favorable as transport cost stays high, monitoring cost stays low, host capability $K_{\mathrm{tech}}(t)$ rises, and the feasibility of successful local instantiation grows. We do not quantify this; we claim only that the inequality is structurally coherent and that it selects *among* the feasible strategies. The optimal-stopping layer is therefore secondary: it chooses within the space that knowledge locality has already shown to be the relevant one. Leading with it instead — treating W as "one option that happens to win under some parameters" — would reintroduce exactly the special-pleading flank that step (5) closes, by making W look like a tunable knob rather than a near-necessity.

4.2 The amplifier: assumption A4

Assumption A4 — that the visitors are narrow rather than uniformly advanced, and so can neither bring nor cheaply operate their own advanced computation and synthesis — enters here, and only here, as an *amplifier*. If A4 holds, waiting externalizes not merely the developmental knowledge that knowledge locality already forces them to source locally, but the *entire* pipeline: design, synthesis, and growth. The economic case for waiting is then maximal.

We flag the role of A4 deliberately, because it is tempting to make it load-bearing and that would be a mistake. The core result — that the visitors face τ_{dev} regardless of their other capabilities — holds for a *uniformly hyper-advanced* visitor, which is precisely why the mechanism inherits the Kent Equation's agent-neutrality. A4 strengthens the *economic* attractiveness of waiting; it is not required for the *logical* necessity of sourcing development locally. A referee who rejects A4 ("why assume they are technologically deficient?") loses the amplifier, not the spine. The dependency A4 describes — that a visitor may rely on the host's computation to design and the host's synthesis to write — sits cleanly in the premise and is fully compatible with the agent-neutral mechanism; it is the *causal story* ("because their development is uneven") that we decline to make load-bearing, since that story is contingent, unfalsifiable, and re-opens the special-pleading objection. We keep the dependency; we demote its explanation.

5 The hybrid requirement and the two gates

The hybrid requirement of step (1) does structural work beyond motivating W: it forces the application to engage *both* technical gates of the Kent decomposition. The hybrid must be *designed and written* — the synthesis gate τ_G, whose output is correct biological information — and it must be *grown* — the developmental gate τ_{dev}, whose output is a body from a sequence. The novel-organism case is exactly where these come apart: synthesizability is computable, but viability for a never-existed target is confirmable only by running development, which is the knowledge-locality argument seen from the other side.

Because the Kent audit establishes that synthesis crosses earlier and development binds ($\tau_G \leq \tau_{\text{dev}}$, so $\tau^* = \tau_{\text{dev}}$), the clock the visitors must watch is $K_{\text{tech}} = F_{\text{dev}}$ *regardless of who supplies design and synthesis.* This yields a clean consequence worth stating, because it separates what A4 changes from what it does not:

Proposition 1 (The amplifier changes what is externalized, not when) *Whether the visitors externalize the entire pipeline (under A4) or only the developmental gate (under knowledge locality alone), the timing of feasibility is identical, because the binding gate is τ_{dev} in both cases and $K_{\text{tech}}(t) = F_{\text{dev}}(t)$ reads the same. A4 alters* how much *of the scaffold is sourced from the host; it does not alter* when *colonization becomes feasible.*

This is the bio-analogue of terraforming named in A5. Terraforming modifies the destination over long timescales so that an unmodified body can live there. ABH inverts the operation: the visitors do not modify the planet to fit themselves; they wait for the planet's civilization to build the capability to fit *them* to the planet, via hybrids grown locally. Read this way, A5's claim that bootstrap colonization is a *primary* mode of expansion does intelligible work — it explains why a visitor would invest decades of patient, low-visibility presence rather than moving on, since on this view Earth is not a curiosity but a colonization project in progress. We note that A5 is a strong assumption: it asserts a general pattern of behavior we cannot observe. It is adopted as a framing that renders sustained patience coherent, not as a fact.

6 Concealment as strategy

ABH must explain an awkward fact that the premise itself supplies: recurrent presence over roughly eight decades, with sustained concealment (A2). The naive explanation — that the visitors are simply reticent — is unfalsifiable and explanatorily idle. Assumption A6 supplies a strategic explanation instead. Premature or unambiguous disclosure risks disrupting the host's developmental trajectory: social destabilization, panic-driven reallocation of resources, regulatory backlash, and the possibility of halting or redirecting the very biotechnology the wait depends on. Concealment is therefore *instrumentally rational* given W — the wait pays only if the scaffold gets built, and clear contact threatens the scaffold.

This is the most valuable of the new assumptions, because it converts concealment from a bug into a *prediction.* If A6 holds, visibility should be *calibrated to the clock*: minimal while K_{tech} is low, when disruption risk is high and there is little to coordinate, and potentially shifting toward managed integration as K_{tech} approaches the threshold, when the disruption cost of contact falls and its coordination value rises. We scope the agency involved as narrowly as possible: A6 posits only that the visitors *manage their own visibility*, which is far more defensible than positing active manipulation of human institutions. We do not propose suppression mechanisms. We claim only that calibrated low-visibility is consistent with an ambiguous observed record — again, consistency, not evidence.

This connects to the social-consequence layer of the Kent paper, which holds that as K_{tech} rises the technical approach drives intensifying social contestation over biotechnology — regulatory conflict, "playing God" rhetoric, public pressure — for endogenous reasons, with causation running from technical approach to social intensification. A6 adds that a waiting visitor would have reason to *modulate* its own visibility against that backdrop. The social phenomena are human-driven; the visitor's contribution is only to time its concealment against them.

7 The clock, and the impossibility of coordinated stopping

Given $V \wedge L \wedge W$, the relevant countdown is the rise of $K_{\text{tech}} = F_{\text{dev}}$, which accelerates as the substrate $M(t)$ matures. This is the ticking clock: Earth's own development drives toward the bootstrap point, and the pressure ABH associates with the approach is driven by a technical trajectory, not by social readiness.

A natural question is whether the host could simply *decline* to cross — whether, knowing the threshold's significance, humanity could stop short of it. We treat this as an auxiliary claim about the *host*, not as a link in the deductive spine, and we resist overstating it. "Impossible to stop" is too strong: individual technologies have in fact been constrained, with reproductive human cloning widely prohibited and biological weapons at least nominally treaty-bound. The defensible claim is narrower and concerns the *aggregate*. The developmental-viability stack is a bundle of dual-use technologies — ex-utero embryo culture, organoid and synthetic-embryo models, laboratory automation, AI for biology, artificial-womb research — each with independent civilian, medical, and economic drivers, and there is no global mechanism capable of coordinating a halt of the bundle. No single technology is unstoppable; the *bundle* is not coordinable. The crossing is, in this sense, overdetermined.

This auxiliary cuts both ways, and the honest paper owns both faces rather than invoking it only where it flatters. On one face, non-coordinability *strengthens* W: the wait is low-risk precisely because the host will build the scaffold whether or not anyone intends to, so the visitor need not act to secure the outcome. On the other face, the same overdetermined co-trending is what makes the human-side signatures non-discriminating (§8–§9): if Earth's trajectory looks the same with or without watching visitors, then almost none of the human-technology evidence can distinguish ABH-true from ABH-false. *The claim that makes the wait safe is the claim that makes the theory hard to test.* Same coin, both faces.

8 Predictions and signatures, and the discrimination problem

ABH generates predictions in three registers. The first is sharp; the others are largely non-discriminating, and we say so.

Technical (sharp). The indicators that matter are *developmental*, not synthetic. Following the Kent reduction, an observer should track: the duration over which embryos can be cultured ex utero; the completeness of synthetic embryo models — specifically whether they advance from forming placenta and yolk sac to forming hearts and brains; and the gestational coverage of artificial-womb technology — whether it extends earlier than the final gestational third. Genome synthesis, which crosses earlier and is not binding, is explicitly *not* the signal; an observer tracking synthesis milestones would systematically misjudge proximity to the threshold. This is ABH's one genuinely specific tell, and it is inherited directly from the mechanism.

Behavioral. If the visitors are transport-limited and waiting, their behavior should resemble monitoring rather than invasion: intermittent appearances, a light cargo footprint, few occupants, interest in markers of host danger and maturity such as nuclear and military infrastructure, non-maximal intervention, long-duration patience, and the calibrated concealment of §6.

Social. As K_{tech} rises, the model expects intensifying contestation over biotechnology, regulatory conflict, religiously inflected "playing God" rhetoric around genetic modification and artificial gestation, and a drift in public framing from "aliens" toward "non-human intelligence."

The discrimination problem. Here we state the central weakness without softening it. The visitors of ABH are *passive* with respect to the human trajectory — they cause none of it. The technical and social signatures above would therefore look *identical* in an alien-free world driven by ordinary technological development. This is the co-trending problem, and for the *alien* layer it is far worse than the version the Kent paper manages on the technical side, because there the binding-gate argument lets empirical content survive in timing and ordering. For ABH's alien layer, almost none of the human-technology evidence discriminates the hypothesis from its negation. Listing more signatures does not help; most of them simply do not discriminate.

9 Falsifiability

The discrimination problem dictates where the only real differential content can live. Since everything on the human side is non-discriminating, the sole prediction that distinguishes ABH-true from ABH-false must reside in the *visitor* layer and in its *coupling to the clock*:

> **Falsification condition.** If ABH holds, visitor-behavior intensity — and, via A6, visitor visibility and disclosure behavior — should *correlate with* $K_{\text{tech}}(t)$: low and concealed while the clock is early, shifting toward managed integration as the clock approaches the threshold. In an ABH-false world there are no visitors to exhibit any such correlation, so the correlation's presence or absence discriminates the hypothesis.

We are blunt about the status of this condition. The data required to evaluate it does not exist, and the premise V is itself unestablished, so the test is at present *unrunnable*. But the *condition is well-formed*: it specifies what observation would discriminate the hypothesis, which is more than an unfalsifiable claim can offer, and offering exactly this — and no more — is the appropriate ambition for a model in this evidentiary position.

There is also a softer but genuinely runnable test, and it rescues ABH from being entirely untethered. ABH's *mechanism* is the Kent Equation, and the Kent Equation makes a checkable ordering claim: that the developmental gate binds while synthesis crosses earlier. That claim is testable against the ongoing biotechnology record. Were synthesis to turn out to bind, or were local instantiation to arrive with no developmental bottleneck, the mechanism ABH depends on would be wrong, and ABH would fall with it. ABH's *premise* is not testable today; its *mechanism* is.

10 Differentiation from nearby hypotheses

ABH overlaps with several speculative frameworks but is reducible to none. Table 1 summarizes the comparison.

11 Limitations and open problems

We catalogue the points a careful referee should press, in roughly descending order of severity.

The premise V is assumed against the current official record. The entire model is conditional on visitation, which NASA and AARO report no confirmed evidence for [2, 3]. Nothing here argues that the premise is true.

Framework	Shared feature	ABH's difference
Zoo hypothesis [4]	Observation without overt contact	Supplies a *logistical* reason for waiting and *dates* it via $K_{\text{tech}}(t)$
Dark forest	Strategic concealment	Concerns colonization *timing*, not hiding for survival
Von Neumann / Bracewell probes	Long-range, patient presence	Local *biological* instantiation rather than machine self-replication
Directed panspermia [5]	Biological seeding	Late-stage *embodiment* within an already-technological host, not early seeding
Ancient astronaut models	Long-term interaction with Earth	Requires no ancient intervention; only possible long-term monitoring
Interdimensional / ultraterrestrial	Non-standard presence	Assumes *transport-limited extraterrestrial* access specifically

Table 1: ABH against nearby frameworks. The strongest single differentiator: ABH explains waiting through colonization *logistics* and *dates* it through the Kent clock, rather than through ethics, secrecy, indifference, or observation alone.

The premise L is assumed and essentially undefended. There is no principled reason to expect any *specific* transport limit; the bell is a conjecture, and the "minimal craft" consistency is weak and over-determined by mundane alternatives. L is the soft spot a hostile reader will dig into first, and we have not removed it — only labeled it honestly.

The human-side signatures are non-discriminating. Because the visitors are passive, the co-trending problem is more severe for ABH's alien layer than for the Kent mechanism, and most of the predicted signatures cannot distinguish ABH from its negation (§8).

The one surviving differential prediction is currently untestable. The K_{tech}-correlated-behavior condition (§9) is well-formed but unrunnable on present data and an unestablished premise.

Assumptions A4–A6 widen the conjunction. Each shapes or strengthens the picture, and each adds attack surface. We have tried to mark which are load-bearing — none of A4–A6 is — and which merely amplify, but a longer conjunction is a more fragile one.

The model borrows its entire technical content from the Kent Equation. ABH contributes a premise and a strategy, not a mechanism, and inherits all of the Kent paper's own limitations, including the unknown required substrate levels and the argued-not-proved status of the developmental-binding claim.

12 Conclusion

The Alien Bootstrap Hypothesis is a disciplined conditional model of delayed extraterrestrial colonization under transport constraints. If extraterrestrial visitation is real, and if transport is constrained, then waiting for the host's developmental-viability threshold is not a tunable preference but a near-necessity, grounded in the locality of developmental knowledge: a novel Earth-adapted organism's viability accrues only by running Earth development, so the visitors face the same gate the host faces, and no advancement elsewhere pre-solves it. Coupling that premise to the Kent Equation converts an indefinite watching into a dated, monitorable forecast whose single discriminating signature is a correlation between visitor behavior and $K_{\mathrm{tech}}(t)$, and whose sharp technical tell is to watch developmental biology rather than genome synthesis.

We have been explicit throughout that the premises are assumed and, by the current official record, unestablished. The reasoning *from* the premises is what the paper offers, not the premises themselves. The exercise earns its place not by being likely true but by being honestly conditional and structurally testable in principle — the mechanism it rests on is checkable even where its premise is not, and its central claim is stated so that some conceivable observation would discredit it. That is a more modest standing than most speculation in this space claims, and a more defensible one.

References

[1] G. Kent, "The Kent Equation: A Time-Indexed Threshold Model for the Onset of Local Biological Self-Instantiation," Rhomega Corp., 2026.

[2] NASA Unidentified Anomalous Phenomena Independent Study Team, *Final Report*, September 14, 2023.

[3] U.S. All-domain Anomaly Resolution Office, *Report on the Historical Record of U.S. Government Involvement with Unidentified Anomalous Phenomena (UAP), Volume 1*, 2024.

[4] J. A. Ball, "The Zoo Hypothesis," *Icarus* **19** (1973) 347–349.

[5] F. H. C. Crick and L. E. Orgel, "Directed Panspermia," *Icarus* **19** (1973) 341–346.